ActInf ModelStream #006.1 ~ "Branching Time Active Inference: the theory

ModelStream | 1:37:07

hello and welcome everyone it is
September 1st 2022.
and we are here in model stream number
6.1
we're going to be discussing branching
time active inference the theory and its
generality
we're going to have a presentation
followed by a discussion so thank you
Ali and Jacob for joining and anyone
else to be adding their questions in the
live chat
without further Ado over to geophilia
Champions
and thanks so much for joining really
appreciate it
hello thank you very much for the kind
introduction uh and thank you for
inviting me to present today I'm very
glad I had this opportunity
so today I will be speaking about
branching time active inference
um basically it's really different
version of the of the the approach the
first reason is Dash passing so second
using Belgian filtering and the third
one is using believe propagation and
allows the model to contain several
observations and several item States
this works has been realized in
collaboration to we've lost Acosta Mike
Josh and Hogwart Bowman
so first of all I want to speak a bit
about the action perception cycle which
is a core ID in active inference

and the second is the agent which is all
of them the environments provide
observation to the agent for example an
image of the environments
and then the agent need to take this
input and perform inference on it and
the goal of the entrance process is to
extract high-level iron states such as
the position of black man in X and Y
or the position of the ghost or whatever
information may be relevant
then based on those States we can
perform planning and action selection
and action selection how it puts the
action to perform maybe the action going
up which is fed into its environment
which produce another observation and
this cycle continues until the Trail
Ends
now that we have the core
there is a core idea of Arctic inference
which is action perception cycle I will
be speaking about active inference in a
bit more depth
so basically active inference is about
an agents which is equips of a model
this agent makes as I said observation
which are represented here at the bottom
of the screen
and those observation depends on the
island States
through the a matrix so that it is a
matrix
provide a distribution of the
observation for each possible Latin
States
we also have the G Vector which contains
the parameter of the prior over the

initial item States
as well as the B Matrix which explain
how the transition of the environment
works so basically it explains how given
a state and an action we get the new
state
at time t plus one
we as I said uh have an action or here
is a policy variable and this action
variable or policy depends on the
Precision parameter which is called
gamma and as we will see influence how
stochastic or deterministic the policy
of the agents will be
so
here we see how the prior over action is
being defined
so it depends as I said on the game
parameter and it is and it is defined as
a soft Max function of minus
the gamma the Precision parameter times
the expected free energy
and the expected free energy for
particular policy is basically a sum of
all future time steps so from t plus 1
which is the first time step into the
future to uppercase team which is a Time
Horizon
and for each time step the expected free
energy is defined as the expected Cost
Plus xiaomi equity
the expected cost is the K Divergence
between as a predictive post trial or
the future observation
and the prior preferences but the prior
preferences defines which observation
the agent want to to observe and this
the predictive posterior defines How

likely its observation is
and so what we want to do is to minimize
the Divergence between the distribution
so that we actually observe what we
learned
okay and the second term is about the
entropy of the likelihood mapping uh
expected under the directional posterior
of a state
so I'm speaking about directional
posterior I will explain what this is in
a minute uh that's that's the definition
of the expected energy risk plus
ambiguity or also called expected cost
per solubility
so I presented the model here's a more
formal definition here we have a joint
over all the variable in the model we
saw that the policy depends upon the
Precision parameter we have a gamma
distribution for the Precision
for each of the tonsor of parameters so
a b and d and we see that s indeed
depends on D
and that's the observation depends on
the state through the a matrix
and 10 thing for the transition mapping
we have a state which depends on the
previous States and the B Matrix as well
as the action being performed in the
environments
okay so now we have the model that what
we want to do is given some observation
we want to be able to compute posterior
beliefs over the lateral variables
in a probabilities in probability we
call that Computing the posterior
distribution

and we do that through a process which
is called inference
we can for example use exact inference
which is based on bias theorem so that's
a complex the posterior is equal to the
likelihood times the prior and then we
normalize using the evidence
and basically the evidence is just
obtained from the numerators by summing
out all the Latin variable X
the problem is that when X is a
continuous random variable as this
summation turns into an integral and we
may not have analytical solution for
this interval
so this method the exact inference can
become intractable because of this
what we do instead when using via snap
inference is that maybe this Trooper
stereo is very complex but we are going
to approximate it using this Q this
viational distribution
and try to make the Divergence between
those two distribution as close as
possible so here we have the trooper
stereo on your writer and here we have
an example of how
the directional posterior may be fit to
the true posterior in red
in the context of active inference the
directionality distribution is defined
as follows so it is a joint distribution
although all the electron variable of
the model and we are doing what we call
the mean field approximation
which means that all the variable within
this distribution are assumed to be
independent thus there is no more

dependency for the pi a b d and Gamma
parameters and we do a slight exception
where the state still depends on the
policy point
okay so that's the definition of the
violational distribution
and now that we have the virational
distribution and the generative model we
can Define the biational Primitive so
the goal of the violational free energy
is to make sure that the
approximate post child so that our
variational distribution remains as
close as possible to the super style and
it is defined as the scale Divergence
between the approximate question and the
generative model
this via style free energy is also
called like the negative evidence lower
bound or elbow in machine learning and
it decomposes into
two intuitive terms so this is the wire
style for energy I decompose it into the
K Divergence
between the approximate and the true
posterior this is the term that will
make the approximate booster will be as
good as possible to the true posterior
and here we have as evidence but which
is a constant with respect to the
viational distribution which we are
optimizing for so yeah really the Via
Sharp Energy is a proxy for the first
term over there
okay so what is
passing is an inference algorithm
basically it is based on what is called
The Mark of blankets so let's suppose we

want to compute the like the virtual
posterior for one specific node in a
graphical model
what national expressing is about is
saying that this node a only depends on
its Mark of blankets
more specifically it depends on the
child of a so here D and C
it depends on the balance of a here f
and g and also in the co balance of a
for example e and B in this picture
and what this macro blanket says that's
really we only need to know the values
of the variable inside the Markov
bracket to be able to perform inference
over a
here is a bit of a bit more formal
um view on this question
so here we have the optimal
virational distribution for one random
variable so this could be for example a
and the query on the right gives us
analytical solution for this uh
posterior and we see that it's only
depending on the Node itself its parents
its children and the cobalance
and that's all we need to know
um okay so why by
saying the message bits comes from the
decomposition of this
iatical solution into messages which are
added together to form the variational
posterior
here we can see the first message which
basically comes from the parents here we
can see one message for each chapter and
what we do and that all those messages
will be added to form the parameter of

the approximate plus tribe
here's a practical example basically we
are trying to perform influence over the
random variable y so we want to compute
the parameter of
um the distribution like the posterior
distribution of a y and the way we go
about doing this is that we send
messages from the parents all the way
through the random variable y
and same thing for the child
and the components and each time we
reach the factor node we combine the
input messages and forward the results
toward y
when we have received all the messages
we just add them together and this
provide us with the parameter of the
violational distribution over y
so this was buyer's Diamond which is
basically the algorithm that we use to
perform in France
now I will be speaking about multicarage
research which is a planning algorithm
that we use to look forward into the
future and estimate the quality of each
product
so before to do that I want just to
introduce the notion of community index
um so here we can see the roots of the
tree which is a state at the present
time step
and then from there we see
[Music]
um
that we have not in the future which are
indexed by sequences of indices for
example here's a sequence of sequence of

size 1 which contains just the action of
the index uh one
so this is a state which after taking
actual one
and then here
we have the states which when performing
action 1 followed by action 2. and this
those indices are called multi-index
because we are they are composed of
several indices
which here represents actions
okay so now that we have this in mind I
can discuss multi-clarative research so
the way Multicultural research is
structure is in four steps first we have
the selection step where we start at the
root node and compute the UCT criteria
so this is just a real value I can
remember for each of the track
and then we select the node which
actually has the highest velocity value
for example maybe it is S1 so the state
with which after taking action
then from this we can compute the
velocity of each children and maybe this
node will be the highest value
okay once we have reached the left node
we can move towards the second phase
which is about expanding the the
children of this node and for each child
what we are going to do is just
simulating some rule outs from this node
I'm Computing the average expected
frequency for this node
once we have this there is a fourth base
which is about updating the expected
finity of the ancestor of uh then that
we just expanded

as well as a number of visits because we
want to explore part of the tree which
have not been explored a lot in the past
so we need to keep track of how many
times each branch has been exploded okay
so that was the algorithm for planning
and I now have introduced always a
background required to discuss the first
approach which is branching time active
infants with bilational massage person
so first we need to define the model it
is basically splitted into two parts the
first one
represents the past and the present and
it is basically a partial observable
micro decision process which is just
offensive word to say that we have
observations which depends on State this
is with a matrix
we have action variables over there and
we have States
as as I said
so electrical machine as I said is
adding like it's parameterized by the a
matrix the transition as usual is
parameterized by the B metric and we
also have here
the Z Vector which defines a prior over
the initial ion States so this is coming
from standard artificial and then the
novelty is that we are expanding the
basically the future
well each time so we okay so we have
this model then we perform active like
um multi-carot research and each time we
want to expand the node what we do that
we are going to add some parts some bits
to the generative model so for example

if we want to add this part of the
generative volume we are going to add
one transition mapping
B this will add a random variable uh is
here it's like S12 and then we will add
the associated observation using the
likelihood map
so this is how we expand the generative
model as we go
mathematically this is defined like this
so we see here a generative model of all
the variable in the model we have the
original prior over the parameters like
of the models so the A B and G matrices
the state at times type zero depends on
D each observation depends on the states
Associated to it and the a matrix
and then we have prior distribution over
actions which depends on some data
parameters for which we have a division
prior
thank you okay and then the state as
usual depend on the previous state and
the previous actions as well as on the B
Matrix which defines the transition
probabilities
so this is upon GP version like so
from GP part of the model and this is a
tree life structure that it expands as
planning is going on
so it is a set of all multi-indexes
which has been expanded in the model for
example if I come back here
we see that there are three multi
indices that have been expanded
so I is equal to
the multi index is one the militant says
two

and the multi-index uh one one
okay and now for each of those military
index we add to the generality model a
transition and likelihood mapping
so that's exactly what we are doing for
each index in the set of loot index
which has been expanded we had
in the future
so now we need to
we need to test this approach in an
environment and for now we will be
testing it inside a maze environment
where we have here is the exit of the
maze here the starting position
and then the prior preferences of the
items will be that the closer we are
from the exits the more the happier the
audience will be
so here we have a distance of zero one
one two three four
um one feature which is really important
is that the prior preferences
fly across fours so here we have we have
a three and here we have four in terms
of distance we don't have like five six
seven eight we have four and what this
produce is a blue cell and what this
blue side blue cell actually is is a
local minimum in which the agent can be
stuck if it is not careful and the way
to avoid this local minimum will be to
be able to plan
final into the future to see that the
rewarding path is actually this one
and notes is the one leading to the
local minimum so this is basically a
challenge that we are adding uh to the
task

and here what we can see so here is
basically an illustration of the tree
which is being expanded and here we have
the table of results we see that as we
increase the number of planning
iteration we go from the agent being
stuck inside the local minimum so never
reaching the limits to an agent which is
basically behaving properly on which
zero sets 100 of the top
next we need to compare how a partial
bunch in time active inference to the
previous like state of the art which is
activated Plus
and basically we designed an environment
in which for the agent to be able to
solve it it has to plan three five and
then eight time step into the future
for active influence what happened is
that it was able to properly solve the
two plus task but then what happened is
that because the number of policy it has
to evaluate into the future grows
exponentially with the number of times
that it has to plan for the last one
which is the biggest crushed
with how
um environments like with our approach
which is the exact same thing and where
we increase once again the number of
planning iterations we see that the
agent becomes able to sort that all the
time
and does not crash because it is able to
explore in a clever fashion the space of
all possible possible policy using
multi-carot research
this was a bit of a non-pherical

comparison between bti and active inputs
now what I want to do in this slide is
to compare them in demo like in terms of
implicity classes which is a lot more
theoretical
so basically here each circle
corresponds to a categorical
distribution of the states which means
that to store one of those Circle we
need to store the number of states so so
a number of parameters which is equal to
the number of State value okay so if we
have a state which takes three values we
need to store three parameters
then for each time step into the future
we need to store one more parameters one
more categorical when it comes to active
inference and we also need to store
um one more category called for each
possible person so the total complexity
class is equal to the number of policies
times the number of time steps until the
time Horizon
times the number of parameters we need
to store for each categorical
distribution
now in the worst case scenario BCI need
to store so the first thing to say is
that bti does not store every single
possible combination it's using the tree
structure of the generative model to
store only one
distribution for the past and present
so first time step in the past and
present and then it's if we expand all
the trade then it's going to ex to store
uh the number of action to the power of
the timer is on minus G

so this is still exponential in terms of
course because of this change over there
but in practice we never expand the
entirety of the trade so what we do is
maybe we will expand this this this and
this but not the two other so in
practice the real complexity class of
this algorithm is a linear in the number
of expansion that we are making
foreign
does not require as much storage space
as standard occupations
now we want I want to speak about the
second approach which is based on
Bayesian filtering
so first what is bias and filtering well
benefit is an inference algorithm which
starts with just a simple generative
model with a state
and an observation we actually know
which observation we are making and we
have we also know what the prior State
and the likelihood is and then we can
for example use
by asteroid to compute the posterior of
the states given some observations
and we just yeah we just complete it
like this
once we have a posterior of a state
we can use it as an empirical prior
so here is our empirical prior and we
also know the transition
probability that lead us to the next
time step
so we can use this information as well
as the action that we're actually
performing in the environment to compute
the predictive posterior over the state

at time step one given the action that
we just made which is u0 and the
observation that we made before
and the way this is done is just by
performing base and prediction as
through the like like the transition
mapping so averaging out the dimension
of s0
and then comes another observation and
we can just use our predictive post
channel that we got from the previous
prediction step to now
um
over S1 according to this new
observation and those two steps of
integrating evidence and then prediction
um through the transition mapping are
going to be iterated as many times as we
as we need to
so this lead us to the second approach
that I want to present today which is
branching time active inference with
Bayesian filtering
so first thing we are not storing any
more uh the past
the past observation on state because
all the information we need is stored
within the beliefs over the initial
State like the Quran timestamp
the state as a strong time step
okay so we have an observation attempty
we can perform the integration of
evidence step to get a belief about this
the state at time step T and then we can
perform forward prediction for each of
the children that we want to expand
for example maybe we'll compute this one
and then this one and so on and so forth

and if multiculties are status to expand
this chart then we will just forward
prediction as well for this one and
expand its Associated observations
so that's the main idea so really the
only difference here is that we don't
even have any more departure observable
lack of decision process for the past
we only have
magnification time step and we are also
changing the inference algorithm from
violation passing to bazel patreon
and here's a more formal definition
of the generality model so we can see
here
the likelihood and prior overstate for
the initial downstep here is the example
team and then for each uh
like
for each future State and observation or
each routine says that we already are
expanded we have
the likelihood mapping and the
transition mapping Associated to it
in terms of performance We compare here
bunching time active inference using
Biogen filtering to the same algorithm
so btai but with various times person
and we see that for the same task one of
them is performing with an order of
magnitude which is about minute scale so
around four minutes between four and
seven minutes one of the other
performers faster between 2 and 11
seconds
so this speed up in performance is
basically uh made possible by the change
of

inference algorithm
but now what we want to do is to be able
to Define more than one observation and
state at each time time point
so what we are going to do this is by
changing once again the insurance
algorithm from
better than filtering to believe
propagation
so what is belief propagation basically
belief propagation is an algorithm that
takes as inputs a function over some
State variables
and this function we know that it
factorizes into a set of n factors which
we call f i
and the question is how can we compute
the marginal distribution like the
national over the marginal of this
function when we marginalize out all the
other random variable except for one SM
okay so we want to marginalize the
distribution of all that SM
and the way I believe propagation is
solving this task is basically by
passing messages
through the computational graph
so in the graph we have two kinds of
nodes we have Factor nodes which
represent factors of the distribution
for example F1 and then we have random
variable
maybe a X1
and we may have several of them so F2
and then X2
okay and maybe
we have a transition mapping between the
two

so this is the second and now what we
want to do is to pass some messages
through the graph so when it comes to a
message from a node X to a factual
so that's because we have one more here
what we are going to do to compute the
output message is just multiply the
input messages
that comes from the other arrows that
you know goes toward this note and we
just multiply all of the input message
and outputs the result
when it comes to a message from
for a factor
we basically are going to
so take the factors for this Associated
to this Factor node
and multiply it by all the incoming
messages so we take all the incoming
messages and multiply them by the
factual associated with this Factor node
and then we marginalize out all the
input Dimensions so that the message at
the exact same shape as the this one the
output as a Target random variables
so this is the marginalization and
speaking of that
we do that for every single messages we
can inside of the photograph and then we
use those messages to compute the
marginal that was the goal of uh of this
algorithm and the way we do that is that
we just take all the input messages
and multiply them together and this
gives us the marginal distribution over
the specific State we wanted to
so this lead us to multimodal on
multi-factor branching time activities

which is the last approach that we have
been developing
before to be able to speak a bit more
about this approach I had to introduce
the notion of temporal styles
so the top work slides is just a set of
states
and observations we have Estates and all
observation
which so this is a plate rotation which
just duplicates a variable all time or
as time
and then we have those dashed line what
those dashed line are doing is just
connecting the observation to a subset
on the states over there so for example
maybe we have an observation one which
depends on stage 1 and state two
and then maybe we have observation too
but this observation 2 only depends on
state two so the reason for which it's a
dashed line is because we can have a
path sparse mapping in uh between the
state and observation we don't have to
have all the possible connection
okay and two slides total product slides
can be connected through the transition
mapping which is these arrows and this
arrows mean is exactly the same but
between two time steps so for example
this uh the state over there can depend
on the state as a previous time step in
arbitrary fashion exactly like the
observation depends on the an arbitrary
subset of the states
but this representation is a bit
unpractical when it comes to presenting
the entire generative model so what we

do is that we just represent these top
wire size
as a square which is called t s t so the
top right size attention and the
background here here is gray because
observation within the top slides are
provided they are actually observed
while in the future the background will
be white uh because the observation are
just not observed
okay so now let's grab this more compact
presentation we can present the
generative model so here we see the
initial time step
and then from there we can expand some
neutral plus size exactly when a
multicult research has asked us to so we
start there maybe we compute the UC
Criterion this is a the highest uh the
node with the highest Criterion so we
are basically asked to expand those
children
and we can do so by just using the
forward prediction
for computing the state here and then
from the state we use forward prediction
once again to predict the future
observation Associated to this top one
slice
foreign
variable once again and it is a product
of the probability of the temporal size
at time T multiplied by all the future
time control size okay so all the top
right size we have already been
expanding during multi-current research
each observation depends uh so in the
initial type of slide so this is the

initial couple of times this observation
depends on the subset of the state
within this topmost size and because we
are at the top of the tree the state at
time step T does not depend on anything
after that for each popular size in the
future we still have the dependency
on uh that we still have the fact that
the observation depends on the state
within the stopwatch slides but we also
have the fact that the state in this
topic slides depends on the state in the
previous like the current top one slides
so for example
the state within this top right size
half balance in inside this top one
slice
okay so this is the way the generative
model is defined
now the way we perform instance is using
what we call the IP IP algorithm so I
stand for inference and P will stand for
prediction
so this slide is about the inference Tab
and the goal is to compute the posterior
over the stage within uh the initial
like the current template slides given
the observations I'm not going to go
through this derivation but you can
pause the video if you are watching it
on YouTube
um
but yeah so basically we do some
derivation and then we obtain this
solution
which just tell us to take the product
over all the electrical mapping with all
the players and then use the build

propagation to actually marginalize out
this function
and that's exactly uh what we are going
to do we use big propagation within the
current timestamp so if I go back here
we have some observation here and we use
build propagation to compute the state
within the initial topographies
so this is the each step the ISTEP which
is the interest of
then we need to perform the p-step each
time the multicult research tells us to
expand a part of the generality model
so once again I'm not going to go
through this derivation but basically
the ID is to compute the posterior over
the state in the next component size
given the observation that we made in
the current template slides
so here we see that balcony once again
we have some kind of summation over all
the parents of the states
in the top five slides that we want to
to compute the posterior form
and this is the transition mapping that
we know and this
is a posterior distribution from the
previous sample size
so basically what we are doing is just
doing forward prediction in this case
taking the expectation of the transition
level
we can do this for the stage
and pause the observations
so to come back to the main picture we
first use the ISTEP to compute the
posterior over the state and the initial
couple of slides

so this is the ISTEP and then we can use
this those posterior distribution to
compute the posterior over the states in
the future so maybe the state in this
double slides and then we can use once
again the key step
to compute the distribution of the
future observations that correspond to
the stop butterflies and we are going to
do that to do that or time step like a
all top of our slides we want to expand
in the future
okay and the next thing that we need to
Define and last for this approach is the
expected free launch so basically is
where we are going to Define this is by
first grouping all the observation into
distant subsets
so basically o i is a set of all
observations in the top right size index
by the muting this ion okay
and now we are going to explain
once we have grouped those observations
into subset the way we Define the
expected free energy is just as a sum
over all possible
uh groups of random variables so each of
those will be iterate over those
and then
into the calendar versions between the
prior
um the predictive post style process
observations
so this is the risk term the color
version between what will happen and
what we want to about
and then we will we will compute the
ambiguity for each of the observation

which is as usual defined as expected
entropy of the likelihood mapping
so this build equation is probably new
for people for for people that have not
created paper
but we can look at one specific case
which makes it more more intuitive and
basically the specific case is just when
each subset corresponds to one
observation in the top five and in this
case the expected frenergy for a
specific processing so for one specific
multi-index
is just the risk for the specific
observation
plus the ambiguity or the specific
observation so we still have ambiguity
plus risk
and now we need
um to present like to test those
approaches together so compare branching
time and second chance with various time
in such testing that isn't filtering and
the last approach which is based on the
propagation
the way we are going to do that is by
using a variance of these five datasets
basically
um we represent
the the environment as a guide
and
so we okay in this environment we have
three different shapes we have additives
us that need to be pulled towards the
bottom right corner of the image and we
have squares and the squares needs to be
pulled on the bottom left corner of the
image

and because there are way too many
positions in X and Y what we do is that
we do some form of State aggregation and
for the eighth first position in the
right left corner
um upper left corner
if the shape is in one of those eight
position we are going to aggregate them
into one state with index 0. and for
this it will be in index one and then
and so on so forth two three four add to
19 for the squares and if it hurts we
will just allocate those States
um the indices between 20 and 39.
and something for instance
so what we are doing is basically
reducing the state space so that some of
those approaches can still do something
because they are not powerful enough to
solve the entire
um State space
so here are the results where we compare
Financial expressing but Bayesian
filtering and the last one which is
based on belief propagation
research passing we had to use a
granularity of 4 which means that the
length like the size of the square
is like a 4x4 set of like a square so as
a cell has a size of four by four
and with this setting we were able to
solve 96 of the times
and the average time for one
um
for one trial is around five seconds
with the base and filtering approach we
were able to go down to the granularity
of two so this times the cells have a

size of two by two
and with this I have with this
granularity the hydrogen becomes able to
solve the task 98 of the time
while
um
but the thing is that because we reduce
the size of the granularity we also
increase the size of the stage space and
this produces an like an increase in
computational times uh which means that
each trailer now requires around 17
seconds to be executed
for the last approach likely we use the
fact that we now know the factorization
of the likelihood and transition mapping
and this allows us to go down all the
way to un like only one the range of one
that's cool so we are now able to
differentiate between every single X and
Y position inside the image
and with this granularity we can solve
the task perfectly and because we can
take advantage of the
like factorization of the distribution
we go a lot faster than all of the
previous approaches and we can solve the
task in around 2.5 seconds
so just I would just want to make one
thing uh very clear here because this
approach directly was able to model
every y position every Exposition every
shapes inside
like you know shapes uh of the display
of environments what is the orientation
like possible orientation of the shape
and or the scale
um so basically each

each shape can have different sizes and
this scale Dimension represent that so
basically this is approach reliable to
to deal with around 700
000 configuration of the state space
now we have presented the results and
show that this approach approaches
daily performance like can lead to very
good performance but now how do we trade
that uh well here I've got a very small
uh code example well directly I'm
retrieving from the environments the a b
c and d matrices
the C Matrix
symmetrics correspond to the
prior preferences of the agent and as I
already said several time a corresponds
to likelihood a b corresponds to the
transition and the is a prior of the
initial States
and then the way we'll go about creating
the bti three agents is just by creating
the top wire slides Builder telling him
that we have one action which is called
a underscore zero and then giving it the
number of value that is that this
instruction can take
then we just add one state for every
single state of our system so the X
position y position shape scale and
orientation and we provide as a second
parameter the pack is the parameter of
the trial over the state so now within
our generative model we have the state
of the system
then we need to add the observation
directly for each of the stage we add
one different one observation which

depends on this state through a matrix
so we provide a matrix and the list of
parents over there
and this basically we add an observation
for each state in the generality model
so before last step is just
um to add transition so basically for
each stage in the system we say what is
the B Matrix that need to be used and
what are the prime the parents for this
text so for example the position in X of
the agent depends on the pronunciation
in X at the previous previous term time
step and the action which is being
performed
foreign
probabilities within the model
and the last step before to build the
top right size is just to Define our
prior preferences and because in the
display data set we need to apply our
preferences about the X Y and the shape
of a blob what we do that we we say here
is one factor so this is one of the X
subsets of observations
and we provide those three Matrix
Associated to that
then we call the the function build
which just returned the top one slice
and we create a PCI three map agent
which uses the stock price slides and we
provide a number of planning iteration
we want the algorithm to use as well as
the exploration constants that Traders
exploration versus exploration
expression exploitation
and this is a graphical user interface
that we have been developing

particularly here we can see the initial
supplys uh if we were in the software we
can click on the standpoint slides and
see the different post style over all
the the state variable
we can also see different information
about those as a stopwatch Styles so the
number of times have been visited on all
this time and then we can use a button
on the on the right on the left to
basically perform a step-by-step
multicarage research so what will happen
in the interface is just we are going to
add the children
and then we will be able to click on
those children to explore those parts of
the trees so and we have information
about the term water slides in the
future
so that's another tool that you can use
to analyze both planning and the beliefs
of the agents
okay so I'm now
um done with presenting the different
approach it's now time for me to
conclude this presentation
so we have seen three different
approaches the first is based on
bioslamic slash passing and obviously
using active entrance and Medical Art
research
the third one is based on business
and the last one is based on the IP
algorithm which is a mixture of belief
propagation and forward prediction
in terms of performance the first so bti
vnp was basically slower like less
performant than the second one which is

once uh less performant than the third
one
okay but even with this uh increase in
terms of performance ability on one
approach to the next there are still
some tasks that we can solve for example
how do we solve image based problem or
which is not clear how we can learn the
structure of the generative model for
now the the modeler has to provide the
description of the model
but it would be very nice if we could
learn it from the data and also
for now we have been basically providing
the actions for like with useful
sequence of actions for example if we go
back to the described compartments we
could imagine a task where each time we
go right it's only pushing the shape one
one position on the right and then one
more and then one more but what we have
been doing to make plan instructable is
just by chunking all those actions
together and maybe executing like eight
action altogether or maybe four action
altogether but this has its limitation
and it will be really nice if we could
run sequences of function like this
automatically
but that's all I wanted to say here are
sugarfrances if you want to to give them
up after the presentation and yeah thank
you
thank you for the presentation indeed
use our reactions
well
very interesting a lot of
material and things for us to discuss

I'll start with just one general
question and then Jacob and Ali looking
forward to your questions
just for some context
how did this team and you come to be
working on this problem were you working
on active inference and interested in
extensions or were you working in a
different adjacent area and came to this
algorithm
okay so maybe one thing that I should
have said is that I'm I've been starting
like I started a PhD at the University
of Kent and
um
basically whole world and Marek which
are two of microbial laboratories on
this project are my supervisors so this
is how we we came up to to work together
just through my PhD and lost to La Costa
is a collaborator that I've been working
with because I've been doing some
presentation at the field which is the
Institute of Neuroscience
um where California is and that is
through a presentation like during a
pronunciation you know to me that he was
um interested in working with me so this
is how I started working with Larson
Acosta and um so in time apply
background I'm coming from a very
computer science like school like follow
into coding and then I arrived at um the
University of Kent where I started to
study about machine learning and this is
where I started to gain some experience
into
reinforcement learning or even active

influence and uh that woman was
initially and Michael just as well was
uh were two of my teachers and robot
woman was already interested in active
inference and one one day when she acted
as a classes I came on scene and saying
well I will be interested to a side
project and that's how of course
everything started and I ended up doing
a PhD after that so
that's it so basically a bit of both I
come from machine learning my address as
well uh long story comes from pure
mathematics and what come from uh like
neuroscience and easy guy which bring in
the marketing from subject to the table
excellent thank you so
I have many more questions but how about
ali first with a question
yeah uh well first of all thanks a lot
for your uh amazing presentation uh I've
really learned a lot uh well I'd like to
make a couple of comments if I may
um and maybe a bunch of questions uh
well as you're well aware active
inference slash fep research has been
going on recently in um basically two
distinct paths uh the theoretical work
geared mostly toward developing the
underlying foundational principles for
example the work of uh Dalton sector by
development Max with Randstad and
colleagues comes to mind and the more
application oriented research
perhaps not unlike the distinction
between theoretical and experimental
physics line of researchers uh but I'm
not sure if you'd agree with me on this

but it seems to me that sadly the
application-oriented research and
especially the work and the various
algorithmic implementations of active
imprints has not gained as much
recognition as it should at least as
compared to the research and theoretical
side I mean the amount of related
published literature is a previous cares
comparatively so in light of this
context
your line of research seems to me a lot
more daring and gains much more
significant so I wanted to congratulate
on that but uh you see I'm a big fan of
unification in science and technology so
my first question is about
your opinion on the possibility of uh
kind of unifying the different unifying
the different algorithmic
implementations of active imprints as an
example you just mentioned the
automatic learning is the structure of
the generative model as a possible
subject for a future and research I'm
not sure if you've seen uh with here at
all's a very recent paper from a couple
of weeks ago namely the learning
generative models for active imprints
using tensor networks if I'm correct
which outlines an interesting physics
inspired approach for that task but it
doesn't include any citations to any of
the branching time active inference the
papers probably because they weren't
aware of your work at the time of their
writing but this works work looks to me
as a pretty good candidate for a

potential integration with btai in order
to overcome some of the limitations you
just mentioned uh or another recent
example uh that would probably be uh
sanish rolls paper deriving time
averaged active influence from control
principles uh which is an attempt to
derive an infinite Horizon
um average uh surprise formulation of
active inference
um so I really liked your comparative
overview of the different variants of
branching time active imprints
especially the Benchmark analyzes and
um I know you described a sophisticated
active inference in your work as a
subset of branching time active interest
but regardless of these are specific
examples I just mentioned I wanted to
ask how do you see the future of btai in
terms of its possible unification with
the other variants of active inference
implementations each with its own pros
and cons in order to overcome some of
the uh some of their limitations without
compromising the advantage messages of
each I mean do you see it as a
possibility that branching time active
imprints will one day subsume all the
other approaches somehow in a truly
integrated kind of framework
so to be honest I don't exactly know I I
the only
like part of the chapter that I've been
exploring is a connection with launching
time active inference has been active in
France and sophisticated entrance and I
haven't been really speak about based on

the pronunciation but I can quickly give
the idea of uh what my worker has been
like what has been the conclusion of
this
and basically
um it is really about
how we back propagate
um what I call the local spectator
energy which is basically as the
expected for energy associated with uh
one node in the future and
so if you back propagate those upward in
the trade following
um
like multi-carot research which
basically comes from bandman's equations
and all this kind of
um literature in reinforcement learning
you will fall into you will underprove
an approach which is very close to
sophisticated imprints basically because
sophisticated inference is also taking
some inspiration into the benman
equations just applying it to the
expected free energy instead of just
having reward uh if you go downward
search that propagates those local costs
towards the future then what you are
effectively doing is just Computing like
the path integral of the expected free
energy and so this will be active
inference just taking the sum of all
future timestart of
of the expected friendship as a great so
this was sophisticated inference and
this will be active influence now
concerning the other approaches that you
mentioned I haven't been reading

um those papers so I consciously State
on that
um but yeah I believe that punching time
actually runs is a very general
framework so it may be the case that
some of them will be related but more
research is not on this this
thank you very much
great comments Jakob do you have any
question
yeah um once again thanks a lot for the
awesome presentation and definite it
definitely explained a lot of uh things
that I didn't understand uh on my uh
reading through through the original
paper
uh I'm wondering uh again going back to
the question of learning the structure
of of the different components of the
generative model
um in in your paper you mentioned you
mentioned uh using
um
deep neural networks as general function
approximators for learning this uh the
state space representation and I'm
wondering whether you have given some
thought of into how neural networks
might fit into this Factory graph
representation of the generative model
and I guess I guess there are perhaps
also two ways to look at learning the
structure of of these different
components one is just the initial step
of uh
in your slide when you showed the kind
of initialization of the model uh
getting the

A and B tensors the the prior the
preferences and and um the prior beliefs
kind of replacing that step with uh with
deep neural networks to learn the
representations but perhaps there's also
another another side to
um where you could dynamically change
the dimensions of these different
components as in perhaps the Asian
receives an observation that wasn't
captured in the likelihood mapping of
the a tensor or perhaps it's a
multi-agent setting where One agent has
affordances the other agent doesn't and
a new transition mapping needs to be
learned through observations uh so um uh
wondering what your thoughts are on this
and
how you think this might be compatible
with branching time active inference
okay so what is the Deep learning
um area is I think pretty interesting
and should be enabling active interns to
scale to a more complicated tasks and
more recently so this paper is not yet
out but I'm working on a deep learning
version of active inference so for now
there is no branching time inside the
picture so there is no more takeout
research
and the reason is because it's already
surprisingly difficult to make it work
just for active influence and
um basically I've been reviewing some of
the paper into Etc and uh and then I
provide my own implementation of a deep
active entrance but for example I speak
I spoke about the display data set and I

was not able to make a deep active
influence work on these approaches on
this on these environments
so yeah and I've been doing a
presentation as a field about this uh
but basically some of the implementation
on the internet contain mistakes
um and yeah some of the paper also
contains some like I mean I'm not sure
if possible paper exact mistake that's
that I don't understand and um for
another zooters I've not been able to to
answer my questions
um so basically what I'm trying to say
that I've been trying to finance deep
active inference and
surprisingly
it's quite difficult to make it work at
least on the despite environment
so there will be a first paper about
analysis analyzing what the Deep neural
networks are actually learning and why
it is failing on this environment
and then
um what I wanted to do is try to apply
this implementation that I hope is
correct
um two different tasks uh most
especially like the target games and
stuff like this and try to find out
whether there are some tasks for which
the expected freemergy and you know is
implementation of the effective
inference can actually solve the task
and the parameter preliminary result
that I have on this
um is that there seems to be some tasks
or which uh deep active influence

is actually performing better than for
example dqm which is a benchmark from
the reinforcement learning literature
and uh yeah that's it's not as
a straightforward basket since
um when I was it's quite challenging to
to implement that was more for large um
deep learning aspects I think it's we
still need to work quite a lot before to
to make something very robust that can
beat uh more standard reinforcement
learning for like benchmarks
and yeah the other opportunities to
structural learning I have not been able
like I have not researched it for now so
we need more time to to think about a
more robust answer to your question
that's yeah that's basically what I had
to say
thank you excellence
one really striking aspect of the
presentation was the analysis of the
computational complexity
so maybe we could return to this because
it's something that we've wondered about
and discussed on a few occasions
yeah you presented the theoretical
complexity classes with a Big O notation
and then also discussed some of the
Practical aspects of the actual like
clock time on a given Hardware wasn't
exactly sure what language or um
Hardware you ran it on but provided the
theoretical complexity class as well as
some runtime provisioning so I was
curious to hear some
thoughts on how does this big O
computational complexity analysis shine

light on different variants of active
inference as well as branchy time active
inference and what real computational
resources were taxed in the analysis was
this a ram overflow that caused the
crash that you referenced earlier is it
a CPU throttling is it paralyzable does
it require temp files like what in
theory is happening with a computational
complexity and the exponential blow up
and then what in practice is going to
facilitate this kind of analysis to
scale
okay so first thing like this complexity
analysis was done in terms of the space
which is required to store all the
parameter of the you know distribution
of the states okay so here we are really
interested in how much space do I need
in order to store all the distribution
of the states like all the posterior
distribution of the states
and but if what happens uh in standard
active inference is that the number of
policy that's will be available to the
agent so let's suppose we have two
actions
we have one here one here okay action
zero action one here at the first time
step there are two actions at the second
time step that we deformed black policy
basically that the agent can
um
can actually perform it can go for zero
zero zero one one zero and one one and
this number of classes will basically be
multiplied by two each time step uh like
for instance on the road right because

each time we can now click each of the
action again for each of the previous
processes
and this exponential growth is quite
problematic
for for example the prior of a policy
social member
um
this definition for the prior of the
policies we see that in order to define
the prior over the policy we need to
confuse the expected free energy for
each of those bonuses
but we need to do that for an
exponential number of them as uh the
timer wasn't applying increases
so this is the first problem
and also
this explanationally exponential
explosion is not limited to the number
of policy because we still remember okay
so maybe for this one I need to go
um and look at the variational
distribution which is used in buyer
style inference but we see that the
number like the biracial posterior of
the states depends on the policy
so for each of the policy we need to
store a distribution of a state
and this is once again a problem because
there's an exponential number of policy
which means that there is an exponential
number of
of
stereo that we need to store
so this is the kind of problem that
appears
um within

um standard Arctic inference the number
of policy is growing exponentially
and we also need to store the
distribution of a state for each time
step
now where launching time active
inference becomes useful is that it uses
a structure like there's a graph
structure to avoid to have to store
every single single possible combination
of uh you know time step versus policy
and so
this those two we see that is growing
linearly is because we just keep in
memory one distribution for each state
in the past and presence
and only when it comes uh the current
time step do we start imagining what's
going to happen in the future
and this growth is still exponential
like if we add if we had to explore
every single possible polar in the
future we will still have an exponential
growth but because we are using
multicarage research
basically we are going to only explore a
small amount of the tree
and this is where the complexity moved
from exponential to linear into the with
respect to the number of expansion of
the model so each time we expand
a new branch in this model we need to
store one more categorical for this uh
future timestamp that's good
and so this is how
uh like we can move from an exponential
[Music]
complexity plus into linear one with

respect to the number of expansion of
the generative model
so that was for the capacity class
in terms of Hardware it was basically
just on my own computer uh some real
critical uh GPU used nothing like this
just CPU basically
very interesting and on the hardware or
on the implementation side where do you
see
packages in python or Julia like Forney
lab and the reactive message passing
Paradigm being developed or do you see
gpus as being relevant like this is the
storage consideration what kinds of
scaling relationships or in theory and
practice how are the operational aspects
of the Computing rather than the space
requirements computed
so the first thing to say is that the
space complexity is also linked to the
time and you know computational
complexity and because for example as I
said when I was speaking about
the
the prior over policy if we have an
exponential number of products you need
to compute an exponential number of
expected free energy on each of those
policies so and same thing for when it
comes to the posterior when we have to
store the Via style posterior and that
there is an exponential number of them
then we also need to compute them so
this will also become extractable in
solution
and um in terms of implementation I know
that's some people have been developing

a phone lab in in Julia I've been
providing my own implementation in
Python
um so yeah those are possibilities in
terms of uh gpus I guess their usage
will be
um
really useful only if the
isographical model allows for
parallelization
for example one case where the gpus are
very useful is for images because each
position in the image can generally be
processed in parallel
so if we had like an alternative model
where we had I don't know likelihood
mapping purchase 4 pixel next to each
others like a patch in the image and we
had to compute the all the posterior for
all the image then there is a very large
potential for parallelization
but if for example a message as a
dependency on a previous message
then there will be um just a part of the
GPU which is just waiting for the input
message to arrive so there is also
um basically limitation on that because
some of the analytical solution requires
some other messages so there are some
dependency throughout the dependency
like the the graphical model by the way
so using GPU yes but probably in
specific generator model for which it's
useful such as image based generative
model or um so yeah discount if we have
something which is very simple then I
think it's not going to benefit a lot
from gpu's complication

excellent
thank you
[Music]
um
Ali again or Jakob if you'd like another
question or I can ask one
yeah I also wanted to ask about the
different
um the possible future implementations
of branching time accident Prince
because
um Daniel and Jacob knows know that I'm
a big Julia fan so I wanted to know if
there's a plan to have a jewelry
implementation a branching time active
inference because I think we already
have a C plus plus and python if I'm
quite right the first branching time
active inference was implemented in C
plus plus and the multimodal one in
Python so what are the future plans for
uh
the other other forms implementation of
btai
so for now I was just planning to just
use Python
um but I guess it should not be too hard
to report it in Julia and just for now I
don't have the usage voice
um let's share that and also for like
future possibility when it comes to
punching time active inference I've been
starting to work on trying to implement
a slant algorithm
um
so this is a simultaneously location and
mapping so that if we create a map of
these environments as we navigate

through it so this is also like a
possibility but within this uh context
basically I was going to go
exponentially that we can have
observation that depends on a very large
number of states and therefore what we
need to have is more like a conditional
probability table which is stored as a
tree so it's biologically
um you can encourage rules with
interconditional probability table for
example you have a school that you have
okay probability of C given a and b
then maybe if a is equal to 1
you want to have
a branching on B and then we have 0 1.
and maybe this is real points okay nine
okay so what this means is that if a is
equal to one and B is equal to zero then
the probability distribution of a c is
going to be 0.1 for the first value on
0.9 inches for the second value and so
basically says the idea is to try to
not represents the entire table but
choose a tree structure to uncut rules
about the dynamic of the world and the
likelihood function as well
and then the challenge is to be able to
perform forward prediction from this
tree and in France also from the stream
and so this is another picture which may
be integrated inside
um which UI in the future
definitely a theme that runs through a
lot of these discussions is representing
objects as trees and then taking the
tree turn or in the forest turn
seriously because the tree structure

allows us to
avoid redundancies and enable some new
types of analyzes Jacob do you have a
question or I can ask one
yeah uh maybe
um continuing on the slam thread I'm
wondering whether uh
you're consider considering the
application of Slam to the
um image classification problem and
perhaps how the image classification
problem needs to be reframed to even fit
this uh well first of all branching time
active inference scheme but overall
active inference scheme because
um it seems that active imprints overall
is
much better suited to these kind of
continuously evolving problems where the
generative process uh changes whenever
um the agent takes actions whereas an
image classification problem
seems to be way more static which uh at
least in the in the machine learning
scheme it's
just input and output and then perhaps
uh some um error
um that gets back propagated through the
network
um so
I'm wondering how you um how you are
thinking about
um
image image classification with uh
active inference and overall just how
images can act
um as input in in
um dynamical environments

So yeah thank you for the impression
once again
um this uh so basically the thing with
like image classification is that we
don't have this stopwatch structure like
you just mentioned which make it quite
difficult for an active in France engine
to be applied to this in some sense it's
a bit like the transition mapping is
is like an identity function in some
stuff like
um yeah it's a bit difficult to think
about how you can bring it but
yeah because each time step could be one
image for example is a classification
um but I think active inference is just
not really well suited to to something
like this for classification I think
they are like just classification models
like you know whatever resonate or
whatever which are much better suited
um
basically if you had to apply active
inference it would have to you have to
have to change the structure of the
model uh to remove that it is a top wall
uh transition and just to have
observation and otherwise you will need
some kind of dynamic or violence like
the Atari games black manuals kind of of
a thing or like the despite environments
and in this case you can model the
temporal dynamics of the environments
and so here active influence really
helps because you can you know think
about action and how they impact the
transition and you can have
so basically an encoder that will

compress the image so you will have
technically an image here you will have
an encoder networks a bit like in a via
channel to encoder that will produce a
parameter so the mean and a variance of
a distribution of the states and then we
will have the decoder over there which
produce another image
from the latent variable basically and
then we will have here like a transition
Networks
which is also a digital networks which
outputs as a mean and variance of a
distribution over the state as a next
time step
and yeah and then here you have another
encoder for the future image and another
decoder for the the future image as well
and these transition networks will have
to take into account basically the
action as well as the states to predict
the next state so that's the kind of
architecture you will need to create a
deep active inference address
and I think it's better suited to
Dynamic environments like attack against
identities or static environments and
this is indeed a lot more complicated to
apply it to
if I could give a few thoughts on image
slam
very fascinating point about static
analyzes
and dynamic analyzes and so
what are some ways that we could pseudo
dynamicize the image classification task
so a few options one of them is
navigation amongst large databases of

images so potentially choosing
informative examples for training in
large empirical image databases or
frames from video or in a dynamic
feedback with prompt engineering for AI
generated images so then it makes it
into a dynamic question response task
that would be using Dynamics at the
level across images but still taking in
the whole image and one other approach
could be building on some of the oculum
motor active inference models of
attention
only taking in a small amount of the
image potentially reducing the state
space or the computational complexity
vastly and then making the Dynamics of
some lower level entity related to
policy selections on eye movements or
attention and then treat that as like
the lower level of the slam and the
classification what kind of image am I
looking at as a higher level of a slam
but the policy is being enacted at the
level of which parts of the image are
being scanned
right
yeah that's indeed a very nice setting
for other ideas
which require registration as a task uh
but as indeed bring you much more like
much better suited to active influence
um so very interesting
I wanted to also ask about
um two
modules or functions that various other
active proposals have had which are
hierarchical nesting of models and

learning
so how do nesting and learning influence
the theoretical and the realized
computational complexity
so I think like I think you actually can
really help but use a computational
complexity of an active infants agents
um for example one idea would be to so I
was speaking about generative model over
images
uh imagine you had like an image and
then so an image can be like it's
millions of possible combination rights
like not even more than that but
probably more than atoms in the universe
but what you could do is like a bit
by imitating the structure of a
convolutional neural networks maybe you
can create over patch of pixels a
generative world that we extract for
example different line patterns so maybe
uh horizontal like or diagonal lines or
horizontal lines and we will have a
first level of your the hierarchy which
will extract those informations and then
you will have you know pattern of
patterns and so on so forth I think you
can create this
um you know as a categorical
distribution but in a VR other model
basically it's a very beginning you have
pixels and then you have small edges and
stuff like this and combination of edges
and all the way up basically to to
having objects
um but this is very complicated like
um in terms of implementation we will
require to like probably use GPU for the

inference process because there will be
very large number of patches so we will
need to speed up the training
but still I think this is a very good
way to reduce the state space of the of
the agents because if you try to to
basically put an image as input of a
standard active influence agents you
will have to have like yeah like more
possible images and numbered or pattern
in the universe even probability with
small images
so hierarchy can really help on that
um and so you had another question right
other than uh this the second aspect was
about learning for example what if we
update our priors as the tree continues
or we want to consider policies on
priors or other types of updating of our
different parameters that might be fixed
in other settings
yeah so one thing to say about
um having learnable prayers is that
in some cases this could go really wrong
so
if you don't like okay imagine you have
observations
and you have States so those are the
observations uh or sorry this is the
states and this will be the observations
so three possible States two possible
observations and if you start with a
Jewish layer prior which is just fully
uniform so maybe the parameters are all
one everywhere
well what's going to happen is that if
you make one of the observation it's
just going to give you as much

um like the entrance process within her
uniform distribution of the states
because the weights within the Matrix
are basically all one so there's for
each observation there is no real
like state which will be more likely to
to basically generate it
which means that the inference process
will have a uniform distribution of a
state and basically
this is a problem because what you will
end up having
is like a a matrix where maybe some
states are more likely but each state is
not more likely to generate different
observations and so basically there is a
failure of learning
because
it's just not able to like if you happen
again so if you remember the way we
update the parameters it's just by
counting the number of states
observation pair that we are being
observing and if the state which is
observed is already is always like one
point like one third for example
like it's a uniform distribution then
it's going to count one third of the
observation for all the state at the
same time and we are not able to
identify which states has been able to
generate this observation because they
are all as likely to generate these
observations and so what is going to
happen that you deviously just cannot
learn which states generate which
observation is just countings in above
time a state has appeared but not you

know with respect to observations
so this is a Visionary case which shows
that adding matrices for example with
derivative and stuff like this can fail
to ruin the dynamic of the environments
and the likelihood of the environment as
well
so maybe having digital networks can you
know avoid this problem uh but yeah it
seems that's a real challenge like
learning the parameters within electric
influence model seems to requires like a
human to cross give it a first drafts
where the
likelihood like that's a cryo is not
uniform when it comes to the Jewish flag
if the model is to be able to run
so this is one of the challenge for
learning in active influence
it's something we've come across in
specifying the state space and what
policies are possible and it it's an
interesting conversation because it
brings us as modelers into engagement
with the model
and helps clarify where are we setting
scaffolds and constraints what
information are we what manifolds are we
placing that agent into that set it up
oh it's rolling downhill within some
super local context even if that local
context is still enormous in its state
space it may still be just the tip of
the iceberg in terms of the total model
structures and that's not even to say we
need to explore the total model
structure in practice but in theory it's
quite important or we might just be

looking where the light is and putting
the rabbit in the Hat making these
models that play out a certain way maybe
even deterministically because they've
been kind of told the secret in the
beginning
um
I have one more um question and then and
then I'll your jaka so
um
you juxtaposed and contrasted three
different approaches which were
variational message passing
Bayesian filtering and belief
propagation
and whether for didactic or pragmatic
use
where do you see these different
approaches as being useful or
specialized
yeah where are they better or where is
one a generalization or a special case
of another
so for example
so the way we structure the model in
bias language especially that we keep
track of the past and each time we get
new observation into the inside of into
the future so let me maybe just go here
um
oh no yeah so okay so invest time is
expressing for example we keep the past
and this is quite interesting because
when you have biochemist passing you can
also have backward messages
which means that as you get a new
observation you will have a network like
the state Associated to it

and what is going to happen is that you
will have a message like this and you
will also have message that go backward
in times and this those messages will
enable you to refine your understanding
of what happened in the past
so this ability to revisit like or you
know update your understanding about the
past is something which is quite uh
specific to the various time massage
passing algorithm and does not appear
for example in base and filterings in
the Basin filtering setting because we
only keep a belief state of of the
current random viable and when we expand
like when it gets a new observation and
a new state Associated to it we are just
going to perform prediction to get the
posterior and then we are going to get
rid of that so we cannot have those
badboard messages
um to address for your beliefs over past
States so we can't really have this kind
of counterfeitual abilities
now with the belief propagation
algorithm that it is very similar in ID
to what is done uh with the belief
propagate in the belief propagation
settings it is just a more scalable
approach which enables one to have a
different item States and different
observation because this
ptibf approach was only restricted to
one observation and one States and if
you have for example the X position
neutral applications of text of as a
displayed environment then you will need
to create one random variable that

corresponds to all the combination of
those X and Y position so maybe it will
be the random viable describing the
position and if we had two value for
each of the you know for the exposition
and the Y position then all the
combination will be like or each of the
value for one times all the password
and and this goes exponentially with a
number of variables so let's suppose we
are now scalable
and that this can variable can take two
additional values then the total number
of combination of those three random
variables will be like eight
eight possible combination basically all
the X and Y position or h of the two
scale possibility maybe scale 1 and
scale two
and this exponential growth becomes
fully magic if you don't have this
ability to have several observation and
several States because you will have an
expansion growth in the number of state
and observation you try to model
and this is where really Alexa the other
approach multi multi-factor and
multimodality is really uh useful to
scale to more complicated approach with
more States and observations
excellent are there Jacob any closing
questions or thoughts
uh well uh you see I came across your
work a few months ago
and it got me truly excited so much that
I read all five of your papers uh is it
is the number five rights I mean uh you
published yeah five papers up to notes

uh because you see more often than not
people see active inference and uh the
free energy Principle as uh basically
speculative uh thinking and Endeavor
without so much pragmatic value in the
real applications so in my opinion uh
your work as I mentioned there's a very
welcome addition to this uh nascent yet
exponentially uh growing feel uh and I
hope to see uh more exciting
developments in the future for branching
time active inference or possibly the
other
uh variants you might come up with in
the future uh and uh I'll definitely
keep following your work from now on
exactly thanks so much for your
joining us today
oh no problem thank you for inviting me
I'm really excited I could present here
so thank you for the invitation
Jacob any final thoughts
yeah uh well this has been a really
great presentation and discussion uh
Ali also uh linked uh your work a couple
of months ago and also uh got me very
excited for uh for the
um future of active inference modeling
and it's a it's a topic that we're
um discussing uh quite a lot in the in
the Institute and
um I think that
this approach to reducing the
computational cost of of uh
of perform of Performing
um active inference in more and more
complex State spaces is
probably the best way to go uh to really

um reach adoption of of these models in
in
different in different domains so yeah
I'm
thank you thank you very much for uh for
joining today and uh I look forward to
reading that paper on on the Deep active
inference deep branching time active
interns sure thank you
well you're welcome back anytime and we
will um certainly be
observing
thank you sure thank you
bye-bye thanks bye
perfect
yeah great
brilliant